

# Augmented Reality for Medical Training in Eastern Africa

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## ABSTRACT

Access to qualified medical care is severely limited in low-resource areas within East Africa. Technologies like Augmented Reality (AR) have been shown to be useful for medical teaching and guidance and may contribute to the training of medical experts in these countries. While still expensive, AR has low infrastructure requirements like cell phones, where ownership has constantly increased over the last few years. However, AR is mainly tested for training and surgical applications in highly developed environments.

In this study, we tested whether AR can also address the needs for healthcare and medical education in East Africa. Our study consisted of a medical need-finding part and a pilot study to address some of these needs with AR. Our study shows that remote teaching via AR led to significantly higher test score gains and more consistent results than in the in-person control group.

**Index Terms:** Augmented Reality, Eastern Africa, Medical education, Hololens 2, Paramedics

## 1 INTRODUCTION

In recent years there has been an increasing interest in Augmented Reality (AR), and Virtual Reality (VR) technologies for applications such as entertainment [4], business, [1] and healthcare [2]. The current price tag of many AR devices make it seem like this technology is only viable for wealthy customers. However, AR and VR have, similar to cellphones, very low infrastructure requirements, making this technology also attractive for low-resource environments, especially for applications that can address crucial needs such as access to qualified healthcare.

Most regions in Eastern Africa have limited access to healthcare. This issue is exacerbated by limited resources and a limited number of medical experts. In this study, we aim to understand whether AR can alleviate some of these problems and play an essential role in medical applications in East Africa.

## 2 METHODS

### 2.1 Interviews

Before designing our study to test whether AR is useful for providing access to qualified healthcare in low-resource settings, we performed literature research and interviews to understand the biggest needs and design our study accordingly. We interviewed a total of 25 students, educators and medical practitioners in Kenya, Rwanda and Tanzania. Recurring issues that came up during the interviews were resource constraints regarding facilities, human resources, and physical space. Interviewees also noted the huge disparity between public and private hospitals within East African countries. Based on the interview sessions and subsequent brainstorming, we established

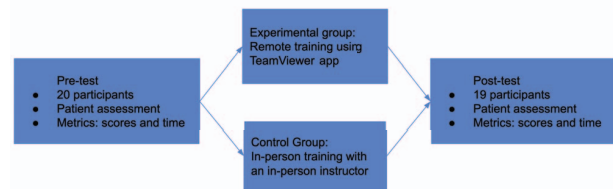


Figure 1: A figure showing a summary of the procedure from the pre-test, the training, and the post-test, and the metrics that were measured.

the need for remote teaching application to distribute knowledge of experts that are otherwise restricted to a few centers and capitals. In the following sections, we designed a pilot study to investigate whether remote teaching using AR is viable in such a low resource setting, whether it can be useful in its current form to support medical education and what are limitations that need to be addressed.

### 2.2 Pilot Study to test AR for remote medical teaching

We conducted the pilot study with twenty undergraduate students from the department of paramedics at Masinde Muliro University of Science and Technology (MMUST). The students consented to participate in the study and to have photos of them taken. The Vice-Chancellor of the School of Nursing, Midwifery, and Paramedics (SONMAPs) gave the department and the team leading the study a go-ahead.

We performed a pre-post-test design to compare the learning gains of a group studying conventionally with an in-person instructor at Stanford University and a group training with the AR device and a remote instructor located physically at MMUST (Fig. 1). The remote instructor used the TeamViewer application to place virtual annotations on the patient's body to support their explanations. TeamViewer is a remote control software that allows an instructor to log in to a computer and connect with students wearing a Hololens device. The instructor can observe the students' field of view and add features such as annotations to the 3D view. The student then sees these annotations projected in their field of view, e.g. text or arrows that point to important anatomy.

## 3 RESULTS

### 3.1 Pre-test

There was no statistically significant difference between the experimental and control groups in pre-test mean score for the chest examination (experimental:  $20.8 \pm 2.8$  points; control:  $20.1 \pm 2.7$ ,  $p=0.65$ ). There was no statistically significant difference in mean procedure completion time (experimental:  $9.6 \pm 2.1$  minutes; control:  $11.6 \pm 3.0$  minutes,  $p = 0.18$ ).

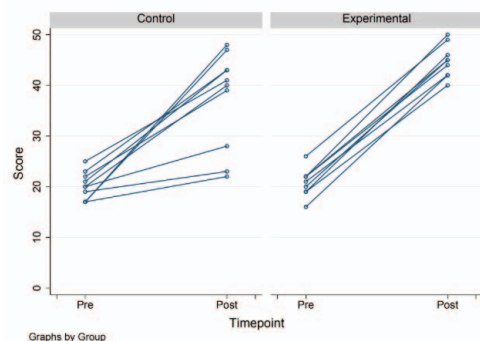


Figure 2: A figure showing the difference in score gains for both the experimental and the control groups.

### 3.2 Post-test

There was a statistically significant difference between the experimental and control groups in post-test mean score for the chest examination (experimental:  $44.8 \pm 2.3$  points; control:  $37.4 \pm 9.6$  points,  $p = 0.04$ ). There was, however, no statistically significant difference in mean procedure completion time (experimental:  $13.4 \pm 5.0$  minutes; control:  $15.9 \pm 5.0$  minutes,  $p = 0.30$ ).

### 3.3 Change in score

Both groups showed a statistically significant improvement in test scores after training, as shown in Fig. 2 ( $p = 0.032$ ). There also was a statistically significant difference in variability of pre-post change in scores between the two groups ( $p = 0.045$  by Mood test due to variance in data). The control group showed a higher variability in pre-post change in score than the experimental group.

### 3.4 Change in time

Neither group showed a statistically significant improvement in mean procedure completion time ( $p = 0.90$ ).

## 4 DISCUSSION

The results showed that the students in the experimental group did significantly better than the control group during the post-test conducted after the training. These results suggest that the AR guided training provided a similar or even better training experience compared to the in-person training. There may be different reasons contributing to the improved training experience for the experimental group.

One reason may be that the virtual annotations provided by the remote instructor via the AR headset improved the confidence and understanding of the students. Prior studies have shown AR technology's role in improving healthcare practitioners' confidence in performing unfamiliar techniques [3]. The different backgrounds of the instructors and the novelty of the AR experience may have also contributed to the students' performance. The remote instructor played back simulated chest sounds during the examination, which may have added to the realism of the teaching experience.

### 4.1 Limitations

#### 4.1.1 Internet Connection

Internet connectivity at MMUST was limited and had a much lower download speed of 5 mbps compared to Stanford University which measured download speeds of 255 mbps. This led to occasional connectivity issues.

#### 4.1.2 Hardware Limitations

One major problem we encountered during training was heating of the Hololens device, which caused it to shut down every fifteen minutes. The Hololens device was also bulky for the students.

#### 4.1.3 Software Limitations

When using the TeamViewer application, the students had to keep their heads still while looking at the patient to allow the instructor to create virtual annotations on the student's camera image. This made the process a bit cumbersome for the students.

#### 4.1.4 Cost

The cost of a Hololens 2 device is \$3,500. Import tax to transport the machine to Kenya is about 35%. The shipping cost is \$350, and only a limited number of organizations ship to Kenya from the US. Shipping and importing one Hololens to Kenya would therefore roughly cost \$5,050, which is still prohibitively expensive.

#### 4.1.5 Sample size

The pilot study was conducted with a small sample size of ten students in the AR and control groups. A future study with a larger sample size is important to ensure that the measured learning improvements using AR are consistent.

#### 4.1.6 Grading system

All students were graded by the same instructor during both pre- and post-test. However, the post-test was done immediately after training and the grading instructor was not blinded to the student's group. In a future study, a blinded student assessment during post-test would be helpful.

#### 4.1.7 Knowledge retention

Our current post-test was performed immediately after training. A follow-up test after a few months would be helpful to test how AR training influences knowledge retention.

## 5 CONCLUSION

The initial interview session with health care practitioners in East Africa supported our findings from the literature reviews on the high need for human, technical, and financial resources to support the health care infrastructure in these regions. In our pilot study we showed that AR technology using remote teaching with the Hololens 2 device can be useful for teaching medical skills remotely via a remote instructor.

To conclude, with decreasing AR prices and increased AR headset availability, telemedicine and long-distance training solutions will become a valuable resource to healthcare professionals in low resource areas.

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